



Nagar YuwakShikshanSanstha's

# Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to RashtasantTukadojiMaharaj Nagpur University)

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NAAC A++

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## Department of Computer Science & Engineering (IoT)

### Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

### Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

### Session 2026-2027

|                                         |                                         |
|-----------------------------------------|-----------------------------------------|
| <b>Vision:</b> Dream of where you want. | <b>Mission:</b> Means to achieve Vision |
|-----------------------------------------|-----------------------------------------|

**Program Educational Objectives of the program (PEO):** (broad statements that describe the professional and career accomplishments)

|      |                             |                                               |                                                                  |
|------|-----------------------------|-----------------------------------------------|------------------------------------------------------------------|
| PEO1 | <b>Preparation</b>          | <b>P: Preparation</b>                         | <b>Pep-CL abbreviation pronounce as Pep-si-IL easy to recall</b> |
| PEO2 | <b>Core Competence</b>      | <b>E: Environment (Learning Environment)</b>  |                                                                  |
| PEO3 | <b>Breadth</b>              | <b>P: Professionalism</b>                     |                                                                  |
| PEO4 | <b>Professionalism</b>      | <b>C: Core Competence</b>                     |                                                                  |
| PEO5 | <b>Learning Environment</b> | <b>L: Breadth (Learning in diverse areas)</b> |                                                                  |

**Program Outcomes (PO):** (statements that describe what a student should be able to do and know by the end of a program)

### Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

**PSO Keywords:** Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

**Integrity:** I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

**Sheikh Wasimuddin**  
**Name of Student**

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|                 |                          |                        |                              |
|-----------------|--------------------------|------------------------|------------------------------|
| <b>Session</b>  | <b>2026-27<br/>(ODD)</b> | <b>Course Name</b>     | <b>Lab : Embeded Systems</b> |
| <b>Semester</b> | <b>7</b>                 | <b>Course Code</b>     | <b>23IOT1706</b>             |
| <b>Roll No</b>  | <b>61</b>                | <b>Name of Student</b> | <b>Sheikh Wasimuddin</b>     |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Practical Number</b>       | <b>07</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Course Outcome</b>         | <b>CO1: Explain the fundamentals of embedded microcontrollers, development environments, and simulation tools.</b><br><b>CO2: Develop embedded applications using Arduino/ESP32 and interface sensors, actuators, and peripherals.</b><br><b>CO3: Implement event-driven and communication-based embedded systems using interrupts, UART, and I2C protocols.</b><br><b>CO4: Design, simulate, and test embedded system applications for real-world applications.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Aim</b>                    | <b>Develop a Smart Water Tank Monitoring &amp; Pump Controller Using FreeRTOS Multitasking</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Theory<br/>(100 words)</b> | <p>A Smart Water Tank Monitoring and Pump Controller using FreeRTOS is an embedded system designed to automatically monitor the water level and control the operation of a water pump. A water-level sensor continuously detects the amount of water present in the tank and provides the information to the microcontroller. Based on the detected level, the controller decides whether the pump should be switched ON or OFF.</p> <p>FreeRTOS is used to manage different operations as separate tasks. The system can have individual tasks for water-level sensing, pump control, display or status monitoring, and user input. The FreeRTOS scheduler manages these tasks according to their priority and timing requirements. This allows multiple operations to run efficiently without one task unnecessarily stopping another.</p> <p>When the water level falls below the minimum limit, the pump is automatically turned ON to fill the tank. When the water reaches the maximum level, the pump is switched OFF to prevent overflow and unnecessary power consumption. Therefore, the system provides automatic water management, improves monitoring, reduces water wastage, and demonstrates the use of real-time multitasking in embedded systems.</p> |



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|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Procedure and Execution<br><br>(100rds) | <ol style="list-style-type: none"><li>1. Connect the ESP32 to the computer and provide power through USB.</li><li>2. Connect the HC-SR04 ultrasonic sensor to the ESP32. Connect <b>TRIG to GPIO 5, ECHO to GPIO 18</b>, VCC to 5 V, and GND to GND.</li><li>3. Connect the relay module to <b>GPIO 19</b> and connect the water pump through the relay contacts.</li><li>4. Connect the green LED to <b>GPIO 21</b>, red LED to <b>GPIO 22</b>, and buzzer to <b>GPIO 23</b>.</li><li>5. Set the tank height to <b>100 cm</b>, with the low-water level at <b>20%</b> and the full-tank level at <b>90%</b>.</li><li>6. Write the ESP32 program using FreeRTOS and create three tasks: <b>Sensor Task, Pump Task, and Alert Task</b>.</li><li>7. Use the Sensor Task to measure the distance using the ultrasonic sensor and convert the distance into the water-level percentage.</li><li>8. Use the Pump Task to check the water level. When the level is below 20%, turn the pump ON. When it reaches 90% or above, turn the pump OFF.</li><li>9. Use the Alert Task to operate the LEDs and buzzer. For low water, the red LED and buzzer are activated. For normal or full conditions, the green LED remains ON.</li><li>10. Create a FreeRTOS mutex to protect the shared water-level and distance variables from simultaneous access by different tasks.</li></ol> <p><b>Execution</b></p> <ol style="list-style-type: none"><li>11. Compile and upload the program to the ESP32.</li><li>12. Open the Serial Monitor and observe the distance and calculated water-level percentage.</li><li>13. Keep the ultrasonic sensor above the tank and change the distance of an object below it to simulate different water levels.</li><li>14. When the simulated water level goes <b>below 20%</b>, the relay activates, the pump turns ON, the red LED glows, and the buzzer gives an alert.</li><li>15. When the water level is between <b>20% and 90%</b>, the system maintains the appropriate pump state and displays</li></ol> |
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the normal condition.

16. When the water level reaches **90% or above**, the relay switches OFF and the pump stops. The green LED indicates that the tank is full.

17. Observe the Serial Monitor to verify messages such as water level, pump status, and alert status.

18. Verify that all three FreeRTOS tasks operate continuously and that the system responds automatically to changes in the simulated water level.

**CODE**

```
//sheikh wasimuddin lab 7 es cseiot 61
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#include "freertos/FreeRTOS.h"
```

```
#include "freertos/task.h"
```

```
#include "freertos/semphr.h"
```

```
#include "driver/gpio.h"
```

```
#include "esp_timer.h"
```

```
#include "esp_rom_sys.h"
```

```
#define TRIG_PIN GPIO_NUM_5
```

```
#define ECHO_PIN GPIO_NUM_18
```

```
#define RELAY_PIN GPIO_NUM_19
```

```
#define GREEN_LED GPIO_NUM_21
```

```
#define RED_LED GPIO_NUM_22
```

```
#define BUZZER_PIN GPIO_NUM_23
```

```
#define TANK_HEIGHT_CM 100.0f
```

```
#define LOW_LEVEL 20.0f
```

```
#define FULL_LEVEL 90.0f
```

```
#define RELAY_ON 0
```

```
#define RELAY_OFF 1
```



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---

```
float waterLevel = 0.0f;
float distance = 100.0f;
bool pumpState = false;

SemaphoreHandle_t dataMutex;

float measure_distance()
{
    gpio_set_level(TRIG_PIN, 0);
    esp_rom_delay_us(5);

    gpio_set_level(TRIG_PIN, 1);
    esp_rom_delay_us(10);
    gpio_set_level(TRIG_PIN, 0);

    int64_t wait_start = esp_timer_get_time();

    while (gpio_get_level(ECHO_PIN) == 0)
    {
        if ((esp_timer_get_time() - wait_start) > 30000)
        {
            return -1.0f;
        }
    }

    int64_t echo_start = esp_timer_get_time();

    while (gpio_get_level(ECHO_PIN) == 1)
    {
        if ((esp_timer_get_time() - echo_start) > 30000)
        {
            return -1.0f;
        }
    }

    int64_t echo_end = esp_timer_get_time();
```



---

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---

```
float duration = (float)(echo_end - echo_start);
float measured_distance = (duration * 0.0343f) / 2.0f;

if (measured_distance < 2.0f || measured_distance >
400.0f)
{
    return -1.0f;
}

return measured_distance;
}

void buzzer_beep()
{
    int frequency = 2000;
    int half_period = 1000000 / (frequency * 2);
    int duration = 200000;
    int cycles = duration / (half_period * 2);

    for (int i = 0; i < cycles; i++)
    {
        gpio_set_level(BUZZER_PIN, 1);
        esp_rom_delay_us(half_period);

        gpio_set_level(BUZZER_PIN, 0);
        esp_rom_delay_us(half_period);
    }

    gpio_set_level(BUZZER_PIN, 0);
}

void sensor_task(void *pvParameters)
{
    while (1)
    {
        float measured_distance = measure_distance();
```



---

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---

```
if (measured_distance >= 0.0f)
{
    float level;

    if (measured_distance >= TANK_HEIGHT_CM)
    {
        level = 0.0f;
    }
    else
    {
        level =
            ((TANK_HEIGHT_CM - measured_distance) /
             TANK_HEIGHT_CM) * 100.0f;
    }

    if (level < 0.0f)
        level = 0.0f;

    if (level > 100.0f)
        level = 100.0f;

    if (xSemaphoreTake(dataMutex,
pdMS_TO_TICKS(100)))
    {
        distance = measured_distance;
        waterLevel = level;

        xSemaphoreGive(dataMutex);
    }

    printf("Distance: %.2f cm | Water Level: %.2f
%%\n",
        measured_distance,
        level);
}
else
{

```



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---

```
        printf("Ultrasonic: No Echo\n");
    }

    vTaskDelay(pdMS_TO_TICKS(1000));
}

void pump_task(void *pvParameters)
{
    while (1)
    {
        float level = 0.0f;

        if (xSemaphoreTake(dataMutex,
pdMS_TO_TICKS(100)))
        {
            level = waterLevel;
            xSemaphoreGive(dataMutex);
        }

        if (level < LOW_LEVEL)
        {
            gpio_set_level(RELAY_PIN, RELAY_ON);
            pumpState = true;

            printf("PUMP: ON\n");
        }
        else if (level >= FULL_LEVEL)
        {
            gpio_set_level(RELAY_PIN, RELAY_OFF);
            pumpState = false;

            printf("PUMP: OFF\n");
        }
        else
        {
            if (pumpState)
```



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---

```
{
    gpio_set_level(RELAY_PIN, RELAY_ON);
}
else
{
    gpio_set_level(RELAY_PIN, RELAY_OFF);
}
}

vTaskDelay(pdMS_TO_TICKS(500));
}

void alert_task(void *pvParameters)
{
    while (1)
    {
        float level = 0.0f;

        if (xSemaphoreTake(dataMutex,
pdMS_TO_TICKS(100)))
        {
            level = waterLevel;
            xSemaphoreGive(dataMutex);
        }

        if (level < LOW_LEVEL)
        {
            gpio_set_level(GREEN_LED, 0);
            gpio_set_level(RED_LED, 1);

            printf("ALERT: LOW WATER\n");

            buzzer_beep();

            vTaskDelay(pdMS_TO_TICKS(300));
        }
    }
}
```



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---

```
else if (level >= FULL_LEVEL)
{
    gpio_set_level(GREEN_LED, 1);
    gpio_set_level(RED_LED, 0);
    gpio_set_level(BUZZER_PIN, 0);

    printf("STATUS: TANK FULL\n");

    vTaskDelay(pdMS_TO_TICKS(500));
}
else
{
    gpio_set_level(GREEN_LED, 1);
    gpio_set_level(RED_LED, 0);
    gpio_set_level(BUZZER_PIN, 0);

    printf("STATUS: NORMAL\n");

    vTaskDelay(pdMS_TO_TICKS(500));
}
}
}

void app_main(void)
{
    gpio_config_t output_config = {
        .pin_bit_mask =
            (1ULL << TRIG_PIN) |
            (1ULL << RELAY_PIN) |
            (1ULL << GREEN_LED) |
            (1ULL << RED_LED) |
            (1ULL << BUZZER_PIN),

        .mode = GPIO_MODE_OUTPUT,
        .pull_up_en = GPIO_PULLUP_DISABLE,
        .pull_down_en = GPIO_PULLDOWN_DISABLE,
        .intr_type = GPIO_INTR_DISABLE
    }
```



---

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---

```
};
```

```
gpio_config(&output_config);
```

```
gpio_config_t echo_config = {  
    .pin_bit_mask = (1ULL << ECHO_PIN),  
    .mode = GPIO_MODE_INPUT,  
    .pull_up_en = GPIO_PULLUP_DISABLE,  
    .pull_down_en = GPIO_PULLDOWN_DISABLE,  
    .intr_type = GPIO_INTR_DISABLE  
};
```

```
gpio_config(&echo_config);
```

```
gpio_set_level(TRIG_PIN, 0);  
gpio_set_level(RELAY_PIN, RELAY_OFF);  
gpio_set_level(GREEN_LED, 0);  
gpio_set_level(RED_LED, 0);  
gpio_set_level(BUZZER_PIN, 0);
```

```
dataMutex = xSemaphoreCreateMutex();
```

```
if (dataMutex == NULL)  
{  
    printf("Mutex creation failed\n");  
    return;  
}
```

```
printf("\n");
```

```
printf("=====\n");  
printf("SMART WATER TANK MONITORING  
SYSTEM\n");  
printf("ESP32 + FreeRTOS\n");  
printf("=====\n");
```



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---

```
xTaskCreate(  
    sensor_task,  
    "Sensor Task",  
    4096,  
    NULL,  
    2,  
    NULL  
);  
  
xTaskCreate(  
    pump_task,  
    "Pump Task",  
    4096,  
    NULL,  
    2,  
    NULL  
);  
  
xTaskCreate(  
    alert_task,  
    "Alert Task",  
    4096,  
    NULL,  
    1,  
    NULL  
);  
  
printf("Sensor Task: CREATED\n");  
printf("Pump Task: CREATED\n");  
printf("Alert Task: CREATED\n");  
printf("FreeRTOS Multitasking Started\n");  
}
```



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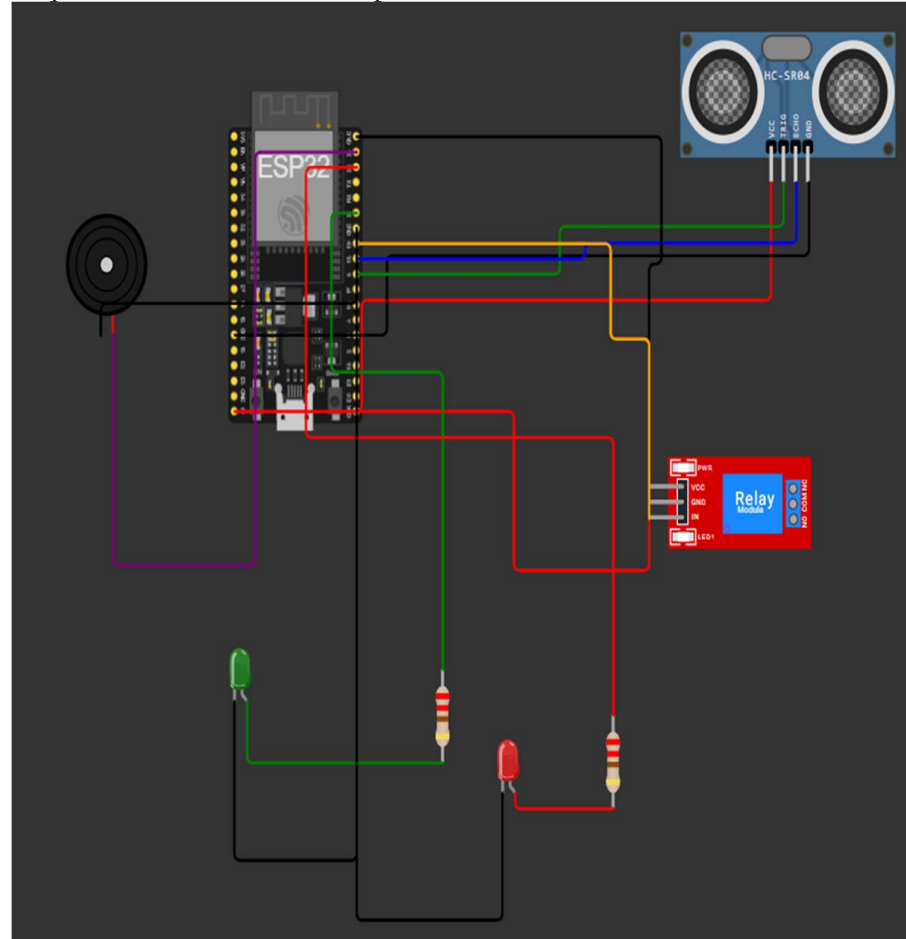
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Stepwise Screenshots with steps:





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WOKWI

SAVE

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61\_Practical7

Docs

main.c

diagram.json

Library Manager

```
1 //sheikh wasimuddin lab 7 es cseiot 01
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include "freertos/FreeRTOS.h"
5 #include "freertos/task.h"
6 #include "freertos/semphr.h"
7 #include "driver/gpio.h"
8 #include "esp_timer.h"
9 #include "esp_rom_sys.h"
10
11 #define TRIG_PIN GPIO_NUM_5
12 #define ECHO_PIN GPIO_NUM_18
13 #define RELAY_PIN GPIO_NUM_19
14 #define GREEN_LED GPIO_NUM_21
15 #define RED_LED GPIO_NUM_22
16 #define BUZZER_PIN GPIO_NUM_23
17
18 #define TANK_HEIGHT_CM 100.0f
19 #define LOW_LEVEL 20.0f
20 #define FULL_LEVEL 90.0f
21
22 #define RELAY_ON 0
23 #define RELAY_OFF 1
24
25 float waterlevel = 0.0f;
26 float distance = 100.0f;
27 bool pumpState = false;
28
29 SemaphoreHandle_t datautex;
30
31 float measure_distance()
32 {
33     gpio_set_level(TRIG_PIN, 0);
34     esp_rom_delay_us(5);
```

Simulation

▶

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■

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00:18.414 100%

Editing Ultrasonic Distance Sensor

Distance: 24cm

STATUS: NORMAL

Distance: 24.16 cm | Water Level: 75.84 %

STATUS: NORMAL

STATUS: NORMAL

Distance: 24.16 cm | Water Level: 75.84 %

STATUS: NORMAL

STATUS: NORMAL

